CO3201 Computer Science Project

Dissertation

Graphical Passwords: Preventing Shoulder Surfing

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DECLARATION

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i. Abstract

This dissertation showcases the development of a multi-modal graphical authentication system aimed at reducing the risks of shoulder surfing attacks partnered with traditional password systems. Ordinary text-based authentication methods are vulnerable to observation attacks, especially in public environments, thereby compromising both user security and privacy.

To address this limitation, the planned system combines multiple graphical and biometric-inspired authentication methods, including gesture input, facial motion recognition, voice authentication, and FaceID-style recognition. These methods are applied within a merged framework that allows users to register and authenticate using one or more selected modalities.

The system is supported by a secure backend that takes care of authentication, template storage, and session management. Modern web technologies are applied, including JavaScript and MediaPipe for real-time input detection, along with Node.js for the backend and PostgreSQL for the database to make sure data is handled securely.

The system was assessed through user testing across all authentication methods. The results indicate that the proposed approach enhances resistance to shoulder surfing while maintaining an intuitive and practical user experience. The findings reveal the effectiveness of combining behavioural and biometric authentication techniques as a viable alternative to ordinary password-based systems.

1. Introduction

Authentication systems are an essential factor in modern digital security, providing mechanisms to verify users before giving access to sensitive information and services. Traditionally, text-based passwords have been the most used authentication method due to their simplicity and ease of implementation. However, despite their popular use, such systems demonstrate significant security challenges. Users regularly create weak or predictable passwords, reuse credentials across different platforms, and remain vulnerable to a wide range of attacks, ultimately undermining overall system security.

One major problem with standard authentication mechanisms is shoulder surfing, an observational attack in which an adversary obtains user credentials by watching their input during authentication. This form of attack is particularly effective in public or shared spaces, where users may unknowingly expose sensitive information. As more people use of mobile devices and authentication in open environments, the risk posed by such attacks has become more prominent, highlighting the weaknesses of traditional password-based systems.

To address these challenges, alternative authentication methods have been reviewed, including graphical and biometric-based methods. Graphical password systems make us of human visual memory and spatial recognition, which have been shown to outdo textual recall in specific contexts. Similarly, biometric approaches such as facial recognition and voice authentication give user-specific characteristics that are more difficult to imitate. However, these methods are not without limitations, including susceptibility to spoofing attacks, surrounding restrictions, and variability in recognition precision.

This project proposes a multi-modal graphical authentication system that integrates multiple authentication techniques, including gesture-based input, facial motion recognition, voice authentication, and Face ID-style facial recognition. By combining multiple modalities, the system adopts a layered security approach, reducing reliance on any single authentication factor and improving resistance to observation-based attacks. This design aims to balance security with usability, ensuring that enhanced protection does not greatly compromise the overall user experience.

The primary aim of this project is to design and implement a secure authentication system that mitigates the risk of shoulder surfing with graphical and biometric techniques. To achieve this aim, the following objectives were defined:

- To develop multiple authentication methods, including gesture-based, facial motion, voice, and facial recognition techniques.
- To integrate these methods into a unified and coherent authentication platform.
- To implement a secure backend system for storing authentication templates and managing user sessions.
- To evaluate the effectiveness, security, and usability of the system in mitigating observation-based attacks.

The system is carried out using modern web technologies, allowing real-time interaction between client-side components and a secure backend infrastructure that processes and stores data. This way demonstrates the feasibility of combining multiple authentication modalities within one system to improve both security and user experience.

The rest of this dissertation is structured as follows. Section 2 provides a review of relevant literature and other existing authentication methods. Chapter 3 gives the system requirements. Chapter 4 delivers the design and implementation of the system. Chapter 5 evaluates the system through testing and analysis. Chapter 6 gives a critical appraisal of the project, and Chapter 7 concludes the dissertation with a summary of findings and directions for future work.

2. Literature Review

2.1 Limitations of Traditional Authentication

Authentication systems have conventionally relied on text-based passwords due to how simple they are and their ease of implementation. However, research has shown that such methods are affected by both security and usability weaknesses. Users usually create weak passwords, reuse credentials across different platforms, and are exposed to attacks such as phishing and brute force. These issues significantly lower the effectiveness of password-based security functions.

Consequently, traditional password-based systems are progressively deemed inadequate for security-critical applications. Their vulnerability is not limited to computational attacks such as brute force or phishing but extends to human-centric threats such as shoulder surfing, which are hard to detect through standard technical controls. This underlines the fundamental limitation of password based systems, as they rely on user behaviour, making them vulnerable to attacks like shoulder surfing, which are difficult to minimise using conventional technical controls.

2.2 Graphical Passwords

Graphical password systems were established as an alternative to text-based passwords, using the human ability to better recognise and remember visual information. According to the ScienceDirect overview [1], these systems replace textual input with images, patterns, or visual interactions, improving memorability and reducing reliance on complex character-based passwords.

Despite these advantages, graphical passwords do not fully address security concerns. However, studies show that graphical passwords, for example recognition-based methods, stay vulnerable to observation attacks, as inputs can be visually captured. While they reduce the likelihood of weak password creation, they often remain vulnerable to observation-based attacks. For example, if a user selects points on an image or draws a visible pattern, an observer may still replicate the input. Therefore, while graphical passwords offer improved usability compared to traditional methods, they do not inherently guarantee stronger security against shoulder surfing.

This indicates that while graphical passwords improve memorability, they do not properly reduce the attack surface in environments where observation threats are common, limiting their effectiveness as a standalone solution.

2.3 Gesture-Based Authentication

Gesture-based authentication extends the concepts of graphical passwords by incorporating dynamic, time-based user input. Unlike static graphical passwords, gesture systems capture both spatial and temporal characteristics, making them more difficult to replicate. Dynamic gesture based systems uses temporal characteristics, making it difficult to replicate than static methods. For example, gesture password capture motion trajectory and timing, forcing an attacker to perform behavioural dimensions simultaneously. Gesture-based systems enhance security by incorporating both spatial and temporal characteristics, increasing the complexity required for successful replication [2].

Similarly, Alsaidi and Najim [2] demonstrate that dynamic hand gesture-based password systems provide increased resistance to replication due to their behavioural nature. Compared to static graphical passwords, gesture-based methods offer a higher level of complexity, as attackers must reproduce not only the shape of the gesture but also its timing and execution.

However, gesture-based authentication is not fully resistant to shoulder surfing. If someone watches your gestures closely or if they're performed in a visible or predictable manner, they may still be observed and imitated. This suggests that while gesture-based systems improve on traditional and graphical passwords, their effectiveness is maximised when combined with additional authentication factors.

To be able to have reliable comparisons of gesture inputs, sequence matching techniques such as Dynamic Time Warping are normally used. It aligns time dependent sequences that could vary in speed and duration, suitable for gesture authentication. This allows the system to deal with natural user input while keeping discrimination between actual and fraudulent attempts. The use of DTW is important in keeping usability and security balanced, as it reduces incorrect rejection caused by natural variation input while keeping resistance to imitation attacks.

Algorithm uses such as Dynamic Time Warping (DTW) significantly enhances the robustness of gesture-based systems by allowing flexible matching of temporal sequences. This reduces false rejection rates caused by natural variation in user input while maintaining resistance to imitation attacks. However, increasing gesture complexity to improve security may negatively impact usability, highlighting a key trade-off between security strength and user experience.

This is supported by Alsaidi and Najim [2], who demonstrate that dynamic gesture-based systems provide improved resistance to imitation due to their reliance on temporal and behavioural characteristics. However, their findings also suggest that gesture complexity can impact usability, reinforcing the need to balance security with user experience.

2.4 Facial Recognition and Biometric Authentication

Biometric authentication methods, such as facial recognition, provide an alternative approach by verifying users based on inherent physical characteristics. AWS documentation [3] describes facial recognition as a process that detects and analyses facial features to identify individuals, offering advantages in convenience and ease of use compared to password-based systems.

Advancements in deep learning have greatly enhanced the reliability of facial recognition technologies. Zhang et al. [7] showcase the effectiveness of convolutional neural networks in face detection and alignment, which serve as the basis for many current facial recognition systems. These methods facilitate precise feature extraction even in challenging environments.

Jain et al. [8] highlight that biometric systems provide strong identity verification by leveraging unique physiological traits, but they also emphasise the importance of secure template storage and resistance to spoofing attacks. This reinforces the need for additional safeguards when implementing facial recognition systems in authentication.

Despite their advantages, biometric systems pose several challenges. They are potentially susceptible to spoofing techniques, such as the use of photos or video, this

shows a critical limitation. While facial features are unique, they are not fully secure without further verification mechanisms. As a result, modern systems often includes liveness detection to improve robustness, which line up with the plan taken for this project through facial motion authentication. environmental factors like lighting and camera quality can also impact performance. Moreover, unlike passwords, biometric data cannot be easily changed if compromised, raising significant security and privacy concerns.

These limitations emphasise the need to incorporate additional security measures, such as liveness detection or multi-factor verification, to reduce susceptibility to spoofing attacks and improve overall system reliability.

2.5 Real Time Facial and Gesture Processing

Modern authentication systems increasingly rely on real-time processing frameworks to enable responsive, interactive user experiences. MediaPipe, developed by Google, provides tools for implementing gesture recognition and facial landmark detection [4][5][6]. These technologies allow for efficient tracking of user input and support the development of advanced authentication mechanisms.

However, while MediaPipe provides robust detection capabilities, it does not inherently address security concerns. Its primary focus is on accurate and efficient processing rather than authentication robustness. As a result, additional layers of logic, such as validation thresholds and behavioural analysis, are required to ensure that these technologies can be effectively applied in secure authentication systems.

While MediaPipe provides highly efficient real-time landmark detection, it functions as a computational tool rather than a security mechanism. Its effectiveness in authentication depends on high level validation logic, such as thresholding and sequence analysis. Without these extra layers, systems relying on landmark detection may be vulnerable to replay or presentation attacks.

2.6 Voice Authentication

Voice authentication is a biometric method that identifies users by unique vocal characteristics. Unlike graphical or gesture-based methods, voice authentication relies on behavioural and physiological features, such as pitch, tone, and speech patterns. This makes it a convenient authentication method, as it does not require visual input or physical interaction with a device.

Voice-based systems function through either speech recognition, which confirms spoken content, or speaker recognition, which verifies the speaker's identity. These technologies are gaining popularity in areas such as virtual assistants and secure entry systems due to their ease of use and accessibility.

The Web Speech API [10] provides a practical framework for implementing voice-based interaction in web applications, enabling real-time speech recognition within browser environments. However, it is primarily designed for usability rather than security-critical authentication, highlighting a limitation when applied to secure systems.

However, voice authentication also has limitations. It may be affected by background noise, microphone quality, and variations in a user's voice due to illness or environmental factors. Voice authentication systems are vulnerable to replay attacks, where recorded audio is used to impersonate a user.

Despite these limitations, voice authentication remains a valuable component of multi-modal systems. When combined with other factors, it contributes to behavioural uniqueness that is difficult to replicate in real time, particularly when dynamic prompts or passphrases are used to mitigate replay attacks. This strengthens the importance of pairing voice authentication with dynamic prompts or other factors to ensure robustness.

2.7 Multi-Modal Authentication

Given the drawbacks of authentication methods on their own, multi-modal authentication has been shown to be a more secure and reliable approach. Rather than relying on a single factor, multi-modal systems utilise multiple authentication techniques to reduce the likelihood of successful attacks.

Each authentication method has distinct strengths and weaknesses, meaning that reliance on a single method creates a point of failure. For example, gesture-based systems improve memorability and give behavioural complexity, while biometric methods such as facial recognition offer convenience and automation. However, when used on their own, each method stays vulnerable to specific weaknesses, such as observation attacks or area limitations.

By having multiple authentication factors combined, multi-modal systems mitigate these weaknesses. An attacker would need to successfully simultaneously compromise multiple independent authentication mechanisms, which greatly increases the computational and observational effort required for unauthorised access. This layered security strategy reduces the chances of both false acceptance and successful attacks.

As reported by Jain et al. [8], combining more than one biometric and behavioural factor increases the difficulty of unauthorised access, as attackers must complete multiple independent mechanisms.

2.8 Summary and Research Gap

The literature demonstrates that while alternative authentication methods improve on traditional password systems, no methods resolve the challenges related with shoulder surfing and observation attacks. Graphical passwords improve usability but still stays vulnerable to observations from outsiders. Gesture-based systems provide behavioural complexity but can still be imitated under certain conditions. Biometric methods offer convenience but raises concerns about spoofing and environmental sensitivity.

This highlights a clear research gap: the need for an authentication system that implements multiple techniques to allow security, usability, and robustness. Existing research mostly focuses on single modality authentication systems, which fails to address the challenges of usability, security, and resistance to observation attacks.

This project points out this gap by designing a multi-modal authentication system that integrates four different authentication methods: gesture-based input, facial motion recognition, voice authentication, and facial identification within a one combined

framework. By putting together these behavioural and biometric factors, the system goals are to reduce relying on any single authentication techniques, also improving resistance to attacks such as shoulder surfing and other observation attacks while keeping usability practical.

3. Design and Architecture

3.1 System Overview

The system created within this project is a multi-modal graphical authentication platform made to increase security against shoulder surfing attacks. Traditional authentication systems rely on one input method, whereas this system allows users to choose and put together multiple authentication techniques, such as gesture-based input, facial motion recognition, voice authentication, and faceID.

The system follows a client-server architecture, with a web-based frontend and a backend server responsible for authentication and managing data. The frontend handles user interaction and real-time input capture, while the backend handles authentication logic, data storage and session control.

By implementing multiple authentication methods within a system. The aim of this design is to provide both flexibility and improved security. Users can select one or more authentication methods, allowing the system to adjust to different usability and security.

This multi-modal design reduces reliance on a single authentication factor, thereby decreasing the system's susceptibility to single-point failures and increasing resistance to both observational and spoofing-based attacks.

3.2 Architecture Design

The system is designed with a modular architecture to ensure scalability, maintainability, and separation of concerns. The architecture is split into three main components:

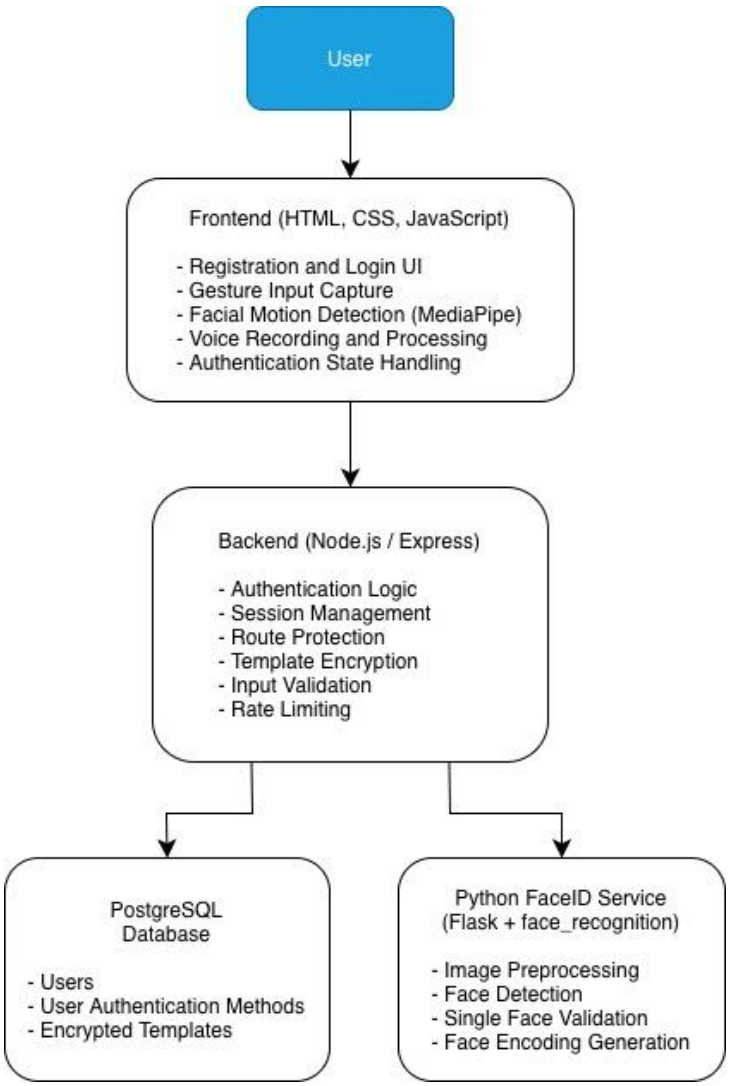
- Frontend Layer:
  The frontend is implemented using HTML, CSS, and JavaScript, providing an interactive user interface for registration and login. It captures user inputs such as gestures, facial movements, voice recordings, and facial images in real time.
- Backend Layer:
  The backend is developed using Node.js and Express, chosen for their asynchronous, event-driven architecture, which is well-suited for handling real-

time input processing and concurrent authentication requests. The backend is responsible for authentication logic, request validation, session management, and secure communication with the database.

- Database Layer:
  A PostgreSQL database is used to store user information and authentication templates. Structured relational storage enables efficient querying and consistency, while support for complex data types allows flexible storage of authentication templates such as gesture sequences and numerical embeddings.

This layered architecture allows each component room to operate independently, improving system reliability and enabling future extensions.

Figure 1: Architecture of the graphical authentication system.



The architecture diagram displays the overall architecture of the authentication system. The frontend captures user inputs, including gestures, facial motion, and voice data, while the backend manages authentication logic, security mechanisms, and communication with persistent storage. A PostgreSQL database stores encrypted authentication templates and user records, while a separate Python-based microservice is used for FaceID processing and facial embedding generation. The modular architecture makes sure loose coupling between components, allowing individual authentication methods to be extended or modified without affecting the overall system.

3.3 Authentication Workflow

The authentication process includes two main phases, registration and login.

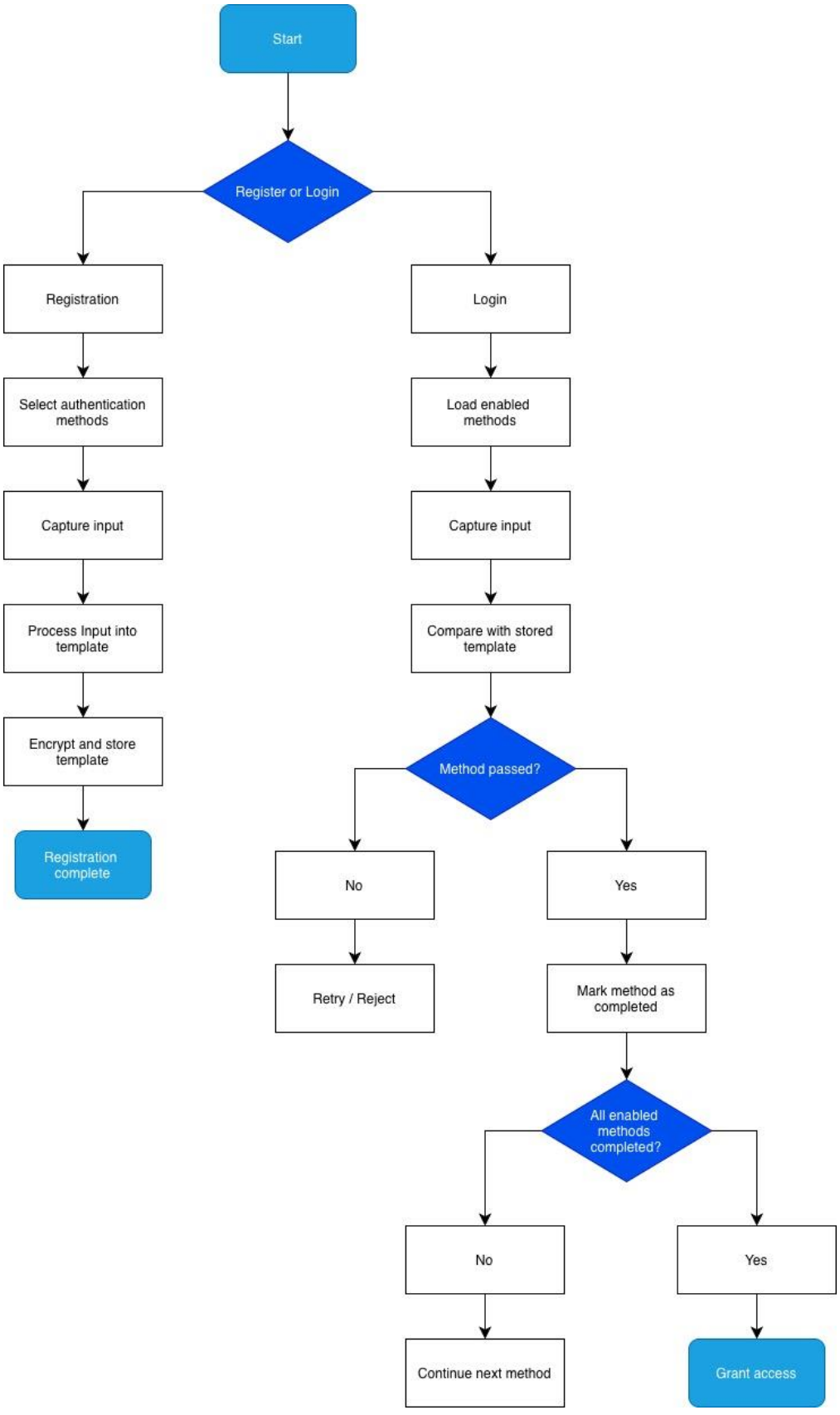
During registration phase, users select one or more authentication methods and provide the input data. For example, a user may record a gesture, perform facial movements, provide a voice sample, or capture facial data. These inputs are then processed and stored as authentication templates in the database.

During the login phase, users must reproduce the authentication methods they have selected. The system captures real-time input and compares it to the stored templates from the database. Access is granted if the input matches the stored data within predefined thresholds.

Matching is performed using predefined thresholds to decide acceptable similarity between the stored templates and live input. These thresholds are critical in balancing false acceptance rates and false rejection rates, ensuring both security and usability.

Traditional systems require all inputs simultaneously, whereas this system allows authentication methods to be completed sequentially in a session. Once a method is verified, it stays valid until the user takes long which will then lead the session to be expired, improving usability without having to compromise security.

Figure 2: Authentication workflow for registration and sequential multi-modal login



The diagram above displays the workflow of the authentication process. In registration, users select the authentication methods they want and provide the necessary input, which is then processed into templates and stored securely by encrypting them. During the login phase, the system loads the enabled methods for the user, so they can confirm each one sequentially. Once a method is completed correctly, the user remains on the login page within the session until all enabled methods have been verified, at which point they will be granted access.

3.4 Authentication Methods

3.4.1 Gesture-Based Authentication

Gesture-based authentication allows users to create a password by performing a simple gesture of their choosing. The system records the gesture as a sequence of spatial coordinates, capturing both the shape and dynamics.

To compare gestures and determine if they match the template, the system uses Dynamic Time Warping, a sequence alignment algorithm that accounts for variations in speed and timing. This allows the system to recognise gestures even when they are not reproduced exactly, improving robustness and user experience.

The use of Dynamic Time Warping lets the system handle inconsistencies, reducing false rejections while maintaining robustness against imitation. This makes it particularly effective for real-world usage, where performing the gesture perfectly would be difficult.

3.4.2 Facial Motion Authentication

Facial motion authentication is based on detecting facial expressions and movements, such as blinking, smiling, or raising eyebrows. This is performed using MediaPipe's facial landmark detection, which tracks points on the user's face in real time.

The system checks predefined facial actions and verifies whether the user performs the needed sequence. This method adds a behavioural layer to authentication, making it more resistant to observation attacks. Unlike static facial recognition, this method

brings liveness detection by requiring dynamic user interaction, reducing the risk of spoofing using static images.

3.4.3 Voice Authentication

Voice authentication is implemented by requiring users to speak a passphrase that they have defined during registration. The system performs both speech recognition and basic voice pattern analysis to verify the input.

In addition to verifying the spoken phrase, the system extracts simple voice features such as frequency patterns and amplitude characteristics. These features are compared to the stored templates to decide whether the input matches the registered user.

This approach combines both speech verification and basic voiceprint analysis, providing a lightweight biometric solution. However, because advanced anti-spoofing mechanisms are not implemented, the system relies on integration with other authentication factors to mitigate repeated attacks.

3.4.4 FaceID Authentication

FaceID authentication is performed using facial embeddings generated by a Python-based microservice. Unlike landmark-based methods from facial motion recognition, which rely on geometric facial features, embedding-based methods represent facial data as 128-dimensional vectors, enabling stronger and more accurate matching.

During the registration phase, when recording for FaceID, the multiple embeddings are captured and averaged to create a stable template. During login, the system computes the Euclidean distance between the live webcam embedding captured and the stored template. Authentication is successful if the distance falls below a predefined threshold.

This approach improves robustness to variations in lighting, pose, and facial expression. However, without advanced liveness detection, the system may remain

vulnerable to high-quality spoofing attacks such as video replay, so I implemented a blinking confirmation function to help reduce them.

3.5 Database Design

The database is designed to securely store user authentication data. Each user has associated records that indicate which authentication methods are enabled, along with the corresponding templates.

The database schema includes tables for information of the user and their selected authentication methods. For example, gesture data is stored as a sequence of coordinates, whereas facial and voice data are stored as processed numerical representations.

Sensitive authentication data is processed into abstract templates rather than storing raw input, reducing the risk of data leakage and improving privacy protection. By structuring the database in this way, the system ensures that authentication data is organised, scalable, and easy to manage.

3.6 Security Considerations

Security is a critical factor in the system design. Authentication templates are processed and stored in a non-reversible format where possible, reducing the risk of direct exposure of credentials. Additionally, encryption protects sensitive data at rest.

The system implements session-based authentication to ensure that users remain authenticated only for a limited duration, reducing the risk of session hijacking. Rate limiting and input validation are also applied at the backend to mitigate brute-force and injection attacks.

The use of multi-factor authentication greatly increases security by reducing reliability to single authentication factors. Even if one method is compromised, an attacker has to

bypass additional mechanisms, increasing the overall complexity of an attack and reducing the likelihood of successful unauthorised access.

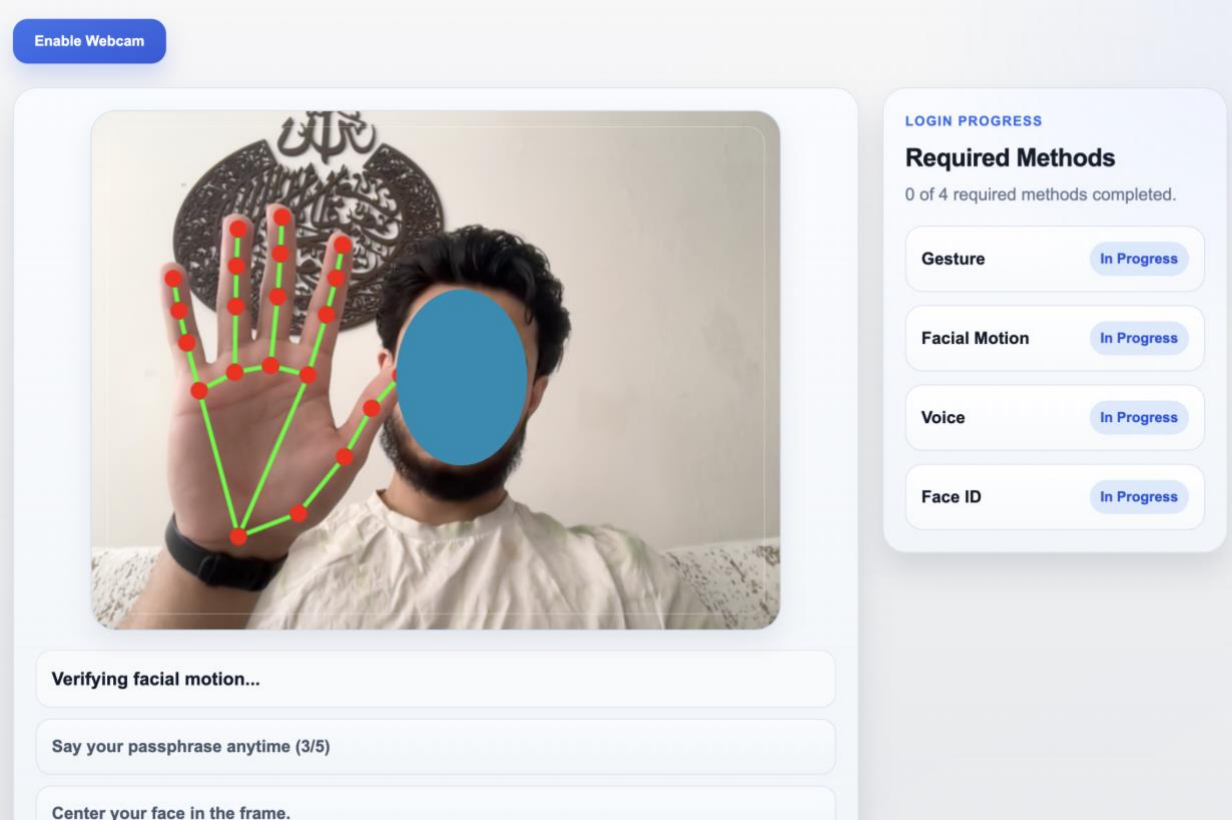
4. Implementation

4.1 Overview

The system was implemented as a web-based application using a combination of frontend and backend technologies. The frontend was developed using HTML, CSS, and JavaScript to provide an interactive user interface, while the backend was implemented using Node.js and Express to handle authentication logic and data management. A PostgreSQL database was used to store user information and authentication templates.

The implementation focuses on real-time processing and low-latency interaction, enabling authentication methods to be captured and verified dynamically without compromising user experience.

Figure 3: User authentication interface showing multiple available authentication methods



4.2 Frontend Implementation

The frontend is responsible for capturing user input and providing feedback during authentication. JavaScript is widely used to handle real-time data processing and interacting with the browser.

MediaPipe was integrated into the frontend to enable gesture recognition and facial landmark detection. The system accesses the user through the webcam and processes video frames when the user starts recording to detect facial features and movements. Canvas elements are used to render visual feedback, such as tracking overlays and gesture paths.

For voice authentication, the Web Speech API is used to capture and process spoken input. Audio data is recorded and analysed within the browser before being sent for verification.

The frontend also manages authentication state, ensuring that once a method is successfully completed, it is retained during the session.

4.3 Backend Implementation

The backend was developed using Node.js with the Express framework [13][14], providing a scalable, event-driven architecture capable of handling concurrent authentication requests and real-time data processing.

Key backend functionalities include:

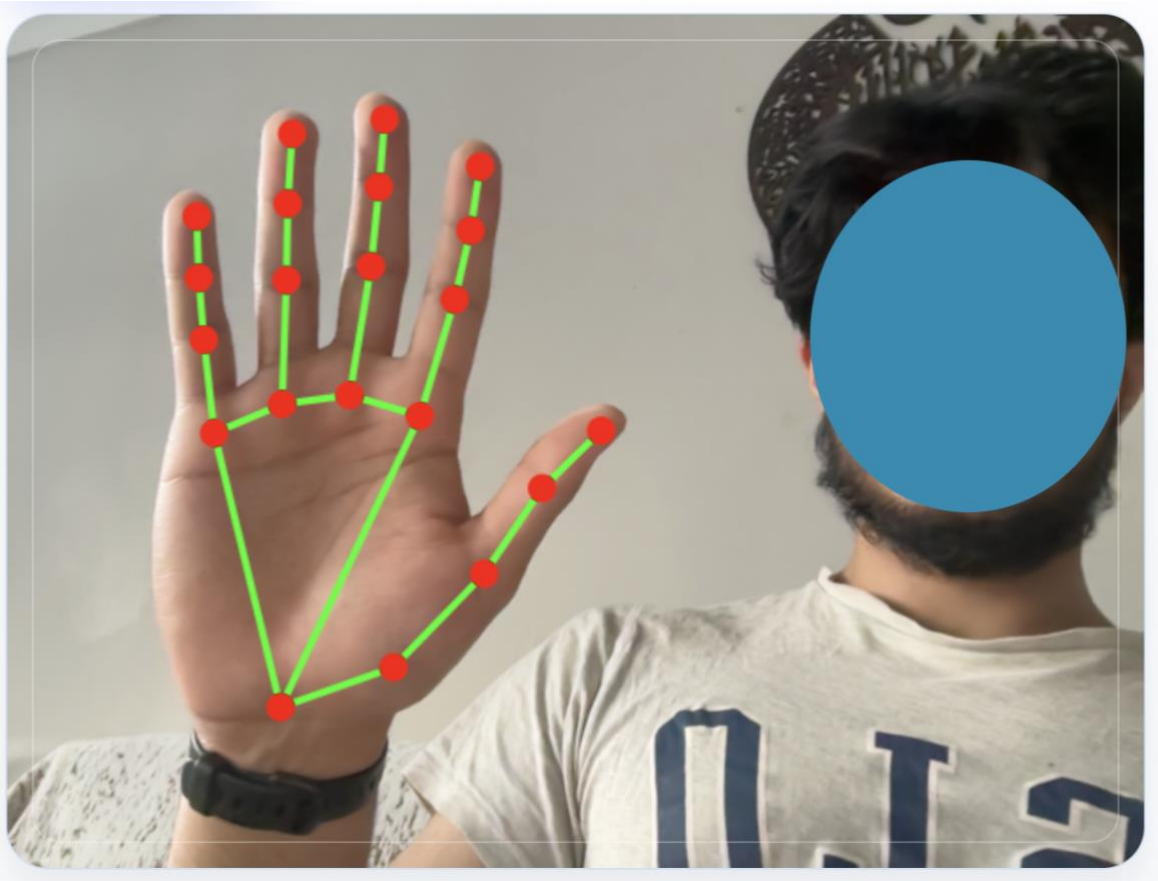
- User registration and login endpoints
- Storage and retrieval of authentication templates
- Validation of authentication inputs
- Session management

Middleware was implemented to enforce access control, ensuring that only authenticated users can access protected routes. Rate limiting and input validation were also applied to help prevent abuse and improve security.

4.4 Gesture Authentication Implementation

Gesture authentication is implemented by capturing the user's hand landmarks in real time and converting them into a sequence of feature vectors. Each frame of input consists of hand landmark coordinates, which are normalised relative to the wrist position and scaled to ensure invariance to hand size, position, and orientation. Also, rotational normalisation is applied to align gestures consistently, to help increase matching accuracy.

Figure 4: Gesture Authentication interface showing real-time hand tracking and input capture



To improve robustness, multiple preprocessing steps are applied to the gesture sequence recording. These include noise reduction through smoothing, removal of redundant frames, and resampling to a fixed set of frames. This is to ensure that the gesture sequences are consistent in length and quality, allowing for a reliable comparison.

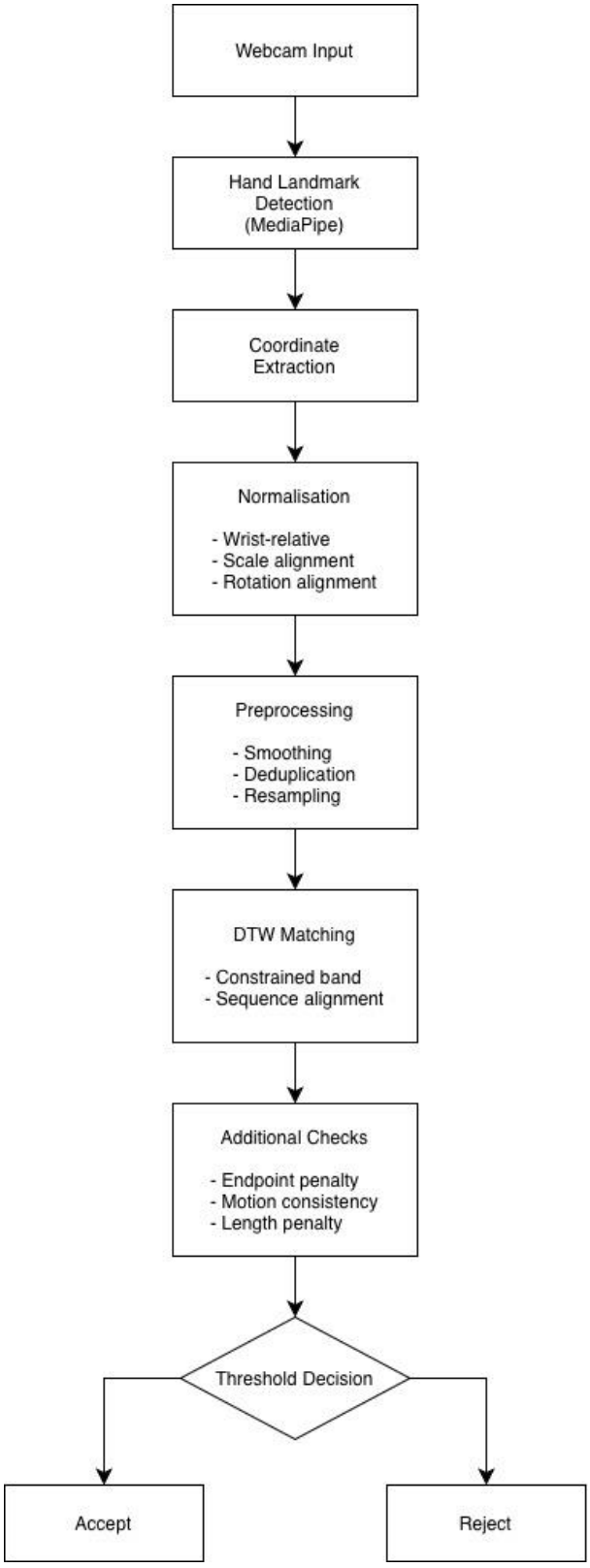
Gesture matching is performed using Dynamic Time Warping, which is a sequence alignment algorithm that calculates the similarity between two-time dependent sequences. DTW gives flexible matching by aligning the sequences that could differ in either speed or timing. A constrained DTW approach is used, including a band limitation to reduce computational complexity and prevent unrealistic alignments.

DTW was selected over simpler distance metrics such as Euclidean distance because it can handle temporal misalignment between sequences. This is important in gesture authentication, as users may perform the same gesture at different speeds. However, DTW introduces higher computational complexity, which is mitigated using constrained banding techniques.

In addition to DTW distance, the system uses multiple validation criteria to help improve accuracy. These include motion consistency analysis, sequence length penalties and checks endpoint alignment. By combining these factors, the system assures that user's gestures are similar in shape and be consistent when executed.

Authentication is successful only when all conditions fall within predefined thresholds, providing a robust and reliable gesture recognition mechanism that is resistant to minor variations and potential spoofing attempts.

Figure 5: Gesture authentication processing pipeline using landmark normalisation, preprocessing, and DTW-based matching



The diagram above displays the authentication pipeline for the gesture method. Using the webcam, it captures the hand landmark data using the technology MediaPipe and converts it into coordinate sequences. The gesture is then normalised to reduce variations in position, scale, and rotation, before being pre-processed through smoothing, resampling and deduplication. The processed sequence is compared with the stored template using constrained DTW, after validation checks such as endpoint, motion, and length analysis. The final threshold decision is then used to decide whether the user performed the gesture correctly and whether it is accepted or not.

4.5 Facial Motion Implementation

Facial motion authentication is performed using real-time facial landmark tracking provided with the use of MediaPipe. The system scans key facial features, such as the eyes, mouth, and eyebrows, to detect specific movements from the facial features and the expression the user makes.

Figure 6: Facial motion authentication detecting user facial expressions in real time



Rather than relying solely on static facial data, this method uses behavioural patterns by requiring users to perform a predefined sequence of facial actions, such as blinking, smiling, or raising their eyebrows. These actions are detected by evaluating changes in facial landmark positions and corresponding expression metrics over time.

Every facial action is detected using a threshold-based method. An example of an action would be blinking, which is detected through monitoring changes in the relative distance between eyelid landmarks, while mouth and eyebrow movements are analysed by the variations in key landmark positions. A movement is only considered valid when it exceeds predefined thresholds, making sure that detections are not triggered by minor or accidental motions.

To help the system become more reliable, it utilises a buffering mechanism that needs consistent detection of an action across several frames before confirming it. So that false positives caused by rapid or unintended movements are reduced.

This way effectively gives a simplified version of liveness detection, ensuring that authentication requires real-time interaction instead of static input, which helps reduce sensitivity to presentation attacks.

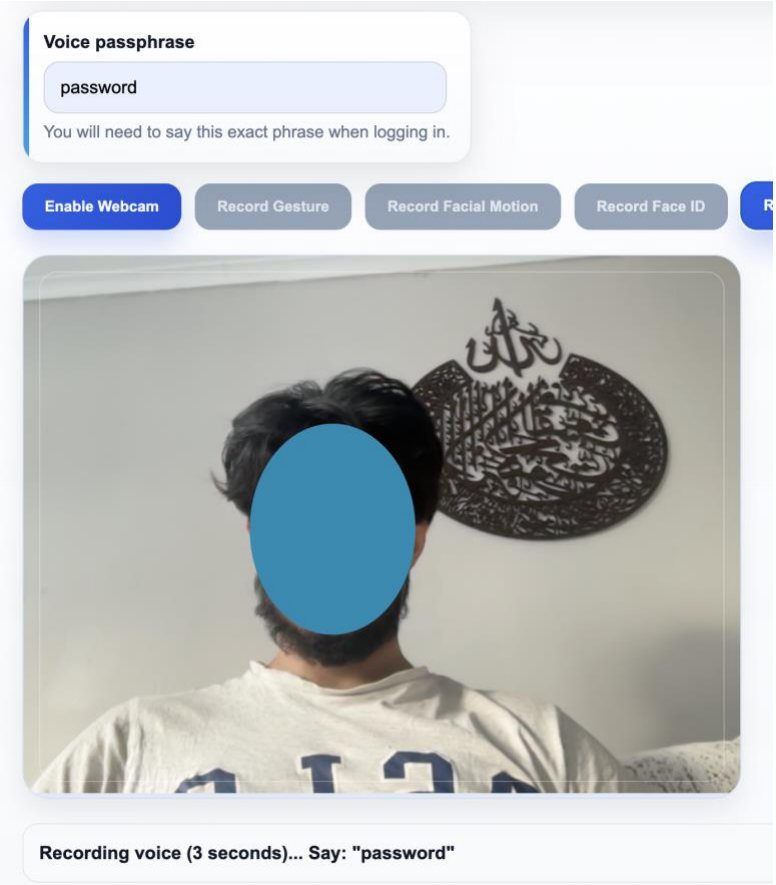
After the system assesses if the user has correctly completed the required sequence of facial actions in the right order. This method is only successful when the entire sequence is performed properly and accurately, which gives an additional layer of security beyond simple facial recognition.

This behavioural approach gives more resistance to spoofing attacks because an attacker would have to replicate both the user's facial expressions and the dynamic sequence of movements in real time.

4.6 Voice Authentication Implementation

Voice authentication is implemented using both speech recognition and basic audio signal analysis. The system checks both the content of the spoken input and the unique characteristics of the user's voice to verify it.

Figure 7: Voice authentication interface showing speech input and passphrase verification



Speech recognition is performed using the Web Speech API, which captures and transcribes the user's spoken passphrase. The recognised text is compared against the expected phrase to ensure correctness. This gives an initial layer of authentication based on knowledge of the user's passphrase.

Additionally, in speech recognition, voiceprint analysis is performed by taking the acoustic features from the recorded audio signal. These features include root-mean-square (RMS) energy, zero-crossing rate (ZCR), and frequency domain components derived from a Fast Fourier Transform (FFT). The extracted features are then combined into a numerical feature vector showing the user's voice characteristics.

To check the user's identity, the system compares the input voiceprint with the stored template using cosine similarity. This similarity measure evaluates the balance between the two feature vectors, allowing the system to decide whether the input

matches the registered user. A predefined threshold is used to decide whether authentication is successful.

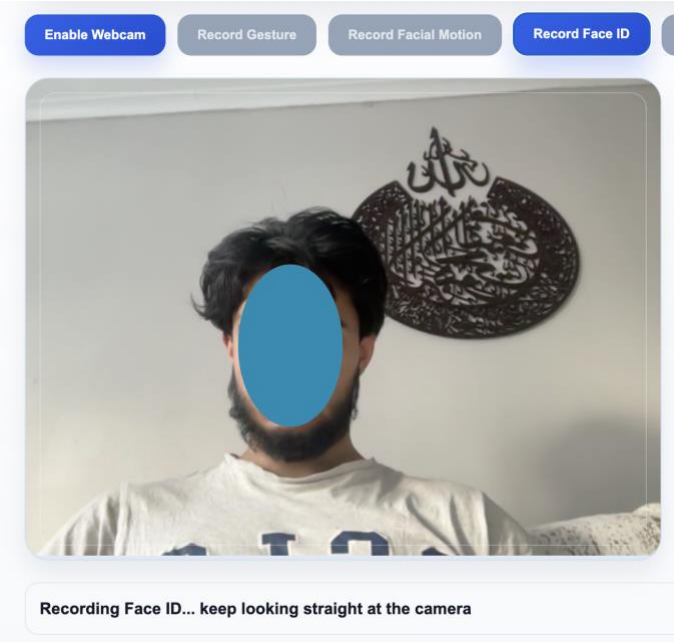
While this approach provides a simple and efficient solution, it is not considered as advanced anti-spoofing techniques such as deep neural speaker verification and therefore relies on multi-modal integration to increase the security for the system.

Speech recognition, combined with voice feature analysis, gives a more secure authentication mechanism than using either method independently. This hybrid approach ensures that both the correctness of the spoken phrase and the uniqueness of the speaker's voice are validated.

4.7 FaceID Implementation

FaceID authentication is created using a Python-based microservice that carries out facial recognition using a deep learning model. This approach separates the computationally intensive face processing logic from the main application, improving scalability.

Figure 8: FaceID authentication process showing facial capture and verification outcome



The face recognition service is developed using Flask and exposes an API endpoint that accepts image input from the main application. When a user gives a facial image, the system sends the image to the Python service, where it is processed and analysed.

The embedding represents facial features in a compact numerical form, allowing effective comparison while maintaining distinguishing information between different users.

When receiving the image, the service performs preprocessing by converting the image into a standard RGB format and turning it into a numerical array so it is able to be processed. Face detection is then performed to check the location of faces within the image. The system has a validation by making sure that only one face is present within the image, rejecting inputs with no faces or multiple faces to stop uncertainty.

When a valid face is detected, a facial embedding is generated using a deep learning-based face recognition model [11]. It is a 128-dimensional numerical vector that transforms the unique features of the user's face. The generated embedding is returned to the application, which uses it for authentication. This way is theoretically aligned with embedding-based models such as FaceNet [9], which represent facial features as high-dimensional vectors for accurate recognition.

In the registration phase, multiple embeddings are captured and averaged to create a stable template. This lowers noise and strengthens robustness against variations such as lighting and minor facial changes. During login phase, a new embedding is produced and compared to the stored template using a distance metric, typically Euclidean distance. Authentication is successful if the distance falls below a predefined threshold.

The implementation of this process within the backend is shown in Figure 16.

Figure 16: FaceID verification using Python-generated embeddings



```
// ---------------------- Face ID Verification ----------------------
app.post('/verify-face-id', upload.single('faceIdImage'), async (req, res) => {
  try {
    const { userId } = req.body;

    if (!userId) {
      return res.status(400).json({
        ok: false,
        message: 'User ID is required'
      });
    }

    if (!req.file) {
      return res.status(400).json({
        ok: false,
        message: 'Face image is required'
      });
    }

    const pythonResult = await encodeFaceWithPython(req.file.buffer, req.file.originalname);
    const liveEmbedding = pythonResult.embedding;

    const faceIdResult = await pool.query(
      'SELECT * FROM face_id_auth WHERE user_id = $1 ORDER BY id DESC LIMIT 1',
      [userId]
    );

    if (faceIdResult.rows.length === 0 || !faceIdResult.rows[0].face_embedding) {
      return res.status(404).json({
        ok: false,
        message: 'No stored Face ID template found'
      });
    }

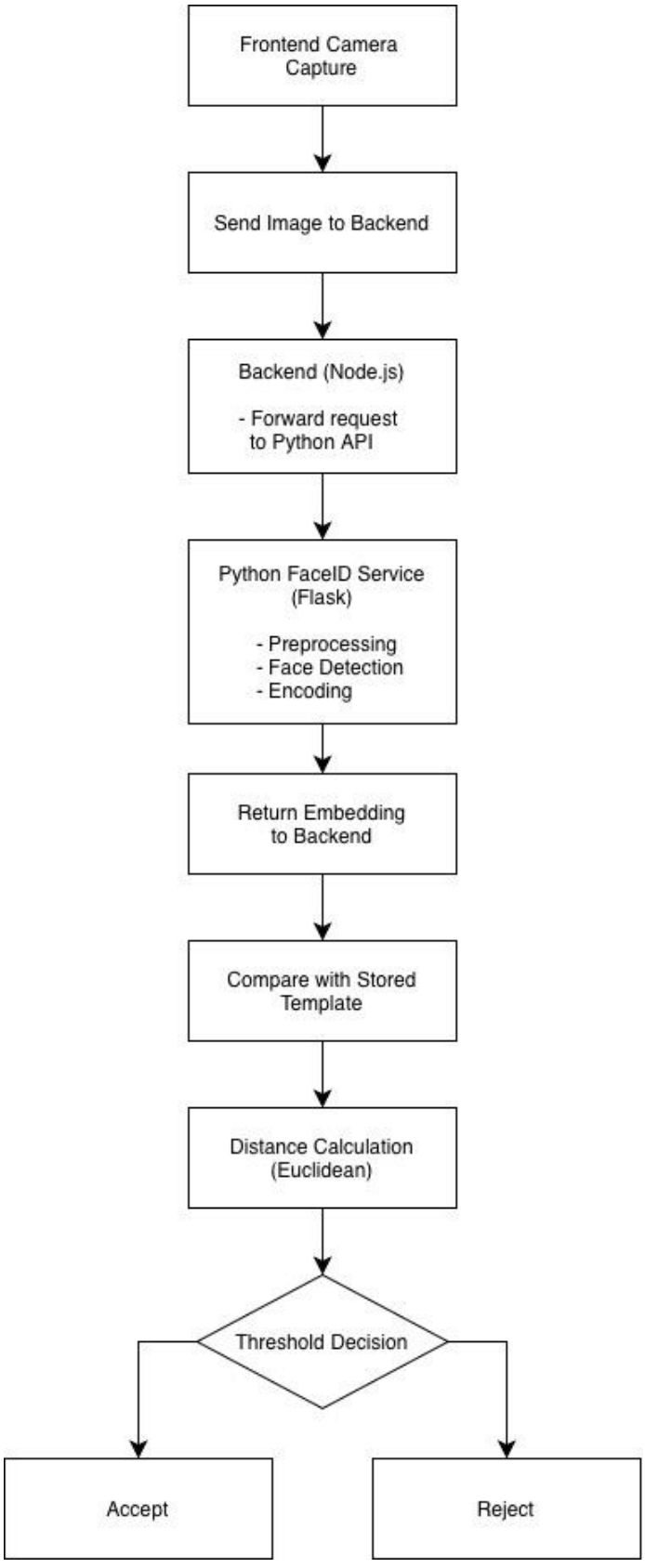
    const rawFaceIdTemplate = faceIdResult.rows[0].face_embedding;
```

As shown in Figure 16, it sends captured facial image to Python-based microservice, where a numerical embedding is generated. The embedding is then compared with stored template which was given from the database. To measure similarity a distance metric is used, and authentication is granted if the value is below a predefined threshold.

This reflects real world biometric authentication systems, where embedding comparisons and threshold values are used to securely verify the user's identity.

This microservice-based approach enables accurate facial recognition while keeping a gap between frontend interaction and backend processing. It also allows for future improvements, such as integrating more advanced models or scaling the service independently.

Figure 9: FaceID process using a Python-based facial recognition microservice



The diagram demonstrates the FaceID process. The frontend uses the webcam to capture an image of the user, which is then given to the backend, forwarding the request to a Python-based facial recognition service. The service performs image pre-processing, detects facial areas, and ensures that only one face is in front of the webcam before a facial embedding is created. The embedding is then returned to the backend and compared with the stored template using a distance metric. A threshold-based decision is then used to decide whether authentication is successful or not.

Although embedding-based methods greatly improve recognition accuracy over the landmark-based method, they are weak to spoofing attacks without dedicated liveness detection mechanisms. This shows a trade-off between implementation complexity and security robustness.

4.8 Database Implementation

The system utilises a PostgreSQL relational database to store user information and authentication data [12]. The database is designed to support multiple authentication methods, allowing each user to have one or more associated authentication templates.

The use of relational design ensures data consistency and integrity, while JSON-compatible structures enable flexible storage of complex authentication templates such as gesture sequences and embeddings.

The database schema includes a primary user table containing basic user information, and an associated table that stores the enabled authentication methods for each user. This relational structure ensures that authentication data is organised efficiently and can be extended to support additional methods in the future.

Authentication templates are stored in structured formats depending on the data type. Gesture inputs are stored as sequences of coordinate data, while voice and facial recognition data are stored as numerical feature vectors. This flexible storage approach allows the system to handle heterogeneous data types within a single framework.

To ensure security, sensitive authentication templates are encrypted before being stored in the database. This prevents raw biometric or behavioural data from being exposed in the event of a data breach. Encryption is implemented using symmetric cryptographic techniques, with additional parameters such as initialisation vectors to enhance security.

Figure 10: Example of encrypted authentication data stored in the PostgreSQL database



The database is accessed through the backend server, which deals with queries and doesn't let the data be revealed to the client. This design keeps a clear gap between data storage and user interaction, improving system integrity.

4.9 Security Implementation

Security is an important part of the system, due to the sensitive nature of authentication data. Layers of security are applied to protect against unauthorised access and common attack vectors.

Authentication templates are first encrypted using symmetric encryption techniques before being stored. This makes sure that sensitive data from the user's selected authentication methods cannot be directly accessed or seen if the database is compromised.

Session-based authentication is implemented using server-side session management. When the user is authenticated, a session is created and kept using cookies. This allows users to complete multiple authentication steps while preventing unauthorised access to protected routes.

To help reduce brute-force and automated attacks, a rate limiting feature is applied to authentication endpoints. This stops the number of requests a user can make within a time period, reducing the risk of repeated login attempts.

Input validation is also used on all requests to stop malicious data from being processed. Cross-Origin Resource Sharing (CORS) policies are configured to control which clients are allowed to interact with the backend.

The blend of encryption, session management, rate limiting, and strict input validation makes the system resilient against common attack vectors, including brute-force attacks, and replay attacks. This multi-layered security approach lines with the best practices in a secure system design, reducing the attack surface while sustaining usability.

5. Testing and Evaluation

5.1 Overview

The system was tested and evaluated through user testing to assess its usability and security. I asked 10 participants to interact with the system and give feedback using a questionnaire document. The evaluation focused on the four authentication methods and was made to assess the usability and system effectiveness, with particular attention to how different authentication methods perform in real-world user conditions. While the study focuses primarily on usability and perceived security, it also considers system responsiveness and consistency during interaction. The main goal of this evaluation was to see how the system effectively meets its objectives of improving usability while improving resistance to shoulder surfing attacks.

Participants were selected in which they volunteered to take part in, through a structured recruitment message, and informed consent was obtained using a participant information sheet highlighting the purpose of the study, data handling procedures, and participant rights.

The evaluation takes both subjective user feedback and observes system behaviour into consideration, giving an assessment of usability, performance, and security characteristics, giving it a good balance in evaluation.

5.2 Usability Evaluation

The results indicate that the usability of the authentication methods differs. Voice authentication was continuously rated as one of the easiest to use, with most participants strongly agreeing that it was simple. Additionally, it was consistently selected as the easiest method overall. This outlines the advantage of voice systems in terms of accessibility and ease of interaction. This suggests that low physical effort plays a significant role in user preference, particularly in authentication systems intended for frequent use.

Figure 11: Shows registration interface used to guide users

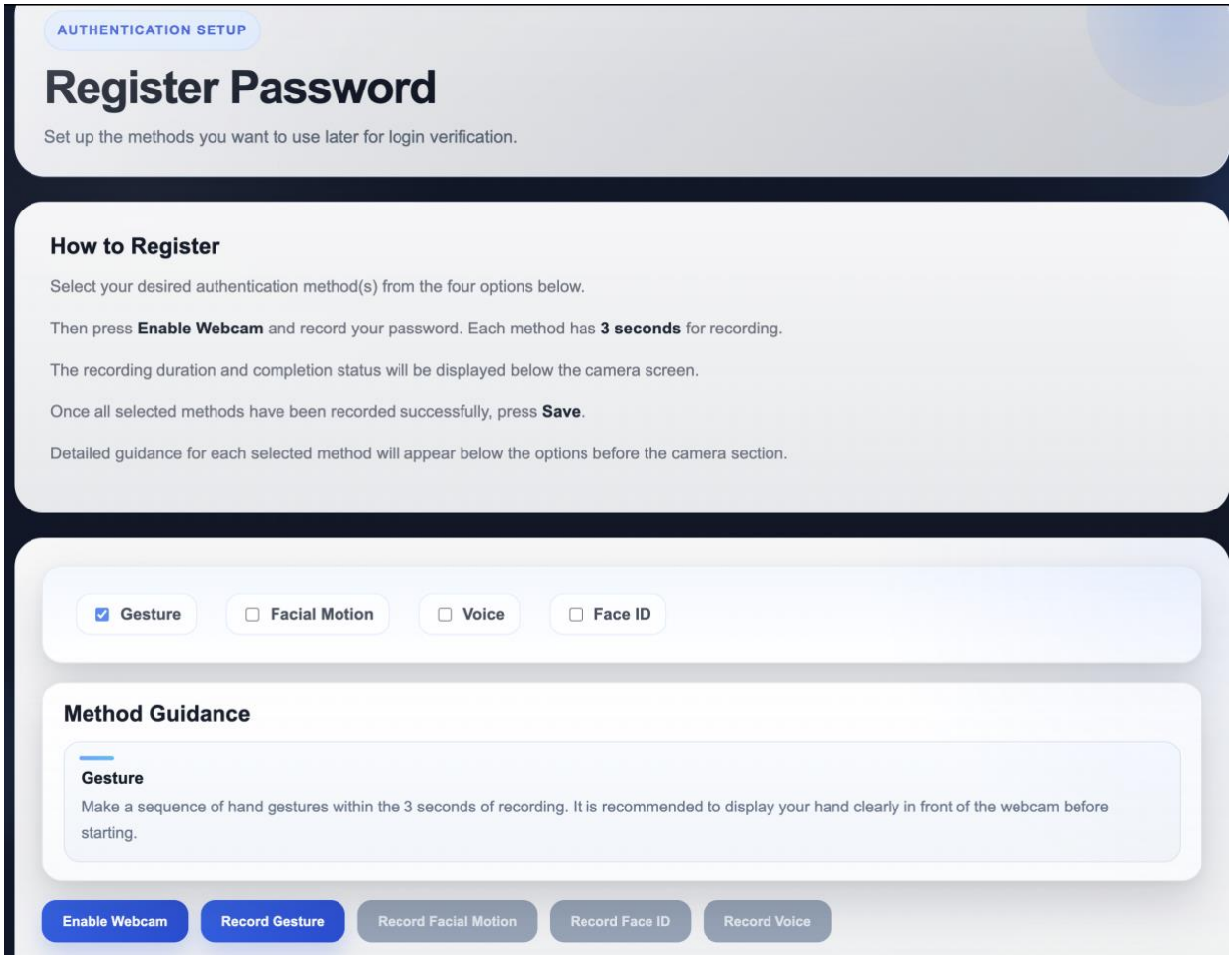


Figure 11 shows the registration interface, which is created to help guide users when setting up their authentication methods. The page includes instructions at the top, explaining the steps needed to register each method. Also, when a user selects a specific method, in this case the gesture authentication, a guidance is displayed to explain how the input should be recorded.

This supports usability by reducing user confusion and supports consistent input, to make it easier for the user to imitate the input during login easily. This contributed to gaining positive usability feedback during evaluation.

Similarly, FaceID was rated highly for ease of use, as it needed minimal effort and was quick to operate. Facial motion authentication also received general positive feedback, although some participants found it slightly less intuitive.

However, gesture-based authentication was often seen as the most difficult method. Participants found it that it was challenging to mimic the gestures accurately to what was registered during login, especially when the user recorded a complex gesture. This reflects the balance between usability and protection, as complex gestures increase resistance to attackers trying to replicate them but reduce ease of use. This exchange is an important consideration in authentication design, where creating complexity improves resistance to imitation but may impact the user's satisfaction and consistency of input.

Table 1 presents a summary of participant responses regarding the perceived usability of each authentication method.

Authentication Method	**Easy (%)**	**Neutral (%)**	**Difficult (%)**
Voice Authentication	70%	20%	10%
FaceID	60%	30%	10%
Facial Motion	50%	30%	20%
Gesture	20%	30%	50%

As shown in Table 1, voice authentication was rated as the easiest method by most participants, with around 70% saying that it was easy to use. FaceID also received positive usability ratings, with 60% of participants indicating positive experiences. In comparison, the gesture authentication was deemed the most difficult method, with 50% thinking it was difficult in consistently mimic their gestures. Facial motion authentication showed more balanced results, with responses distributed across all categories.

This suggests that user preference is influenced by interaction effort where methods need minimal input are preferred. On the other hand, this does not mean it indicates better security, highlighting a gap between usability and robustness.

5.3 Security Evaluation

Participants generally see the system as more secure than traditional passwords, with the majority of the responses being in agreement or strong agreement. However, perceived security does not reflect actual resistance to attack, as users may associate biometric systems with higher security without knowing their potential weaknesses. This suggests that multi-modal authentication increases the confidence of the users in system security.

It is important to know the difference between perceived security and actual security. While participants may link biometric methods such as FaceID with strong protection, these systems can still be vulnerable to specific attack vectors, including spoofing and replay attacks.

FaceID authentication was naturally seen as one of the most secure methods by participants, as it is highly likely because of the biometric data and the user interaction is kept to a minimum. This aligns with user expectations, as facial recognition is commonly associated with devices such as Apple iPhones.

This indicates users put their trust in biometric authentications as they may be influenced by being familiar to commercial technologies, instead of fully understanding potential vulnerabilities.

From a technical view, embedding-based facial recognition improves matching accuracy and robustness compared to a simple landmark-based method. However, without advanced liveness detection mechanisms, systems may still be vulnerable to spoofing techniques.

To enhance robustness, the system utilises a basic form of liveness detection by making it mandatory for the user to blink during authentication. This helps prevent simple spoofing attempts using images, as real time movement is needed.

However, while this improves security, the evaluation was not tested against spoofing attacks. Therefore, although FaceID was considered secure, its security against more advanced attacks was not evaluated and assessed in the study.

Gesture-based authentication also contributes to security through its behavioural nature, which requires users to reproduce dynamic patterns that are difficult to observe and imitate. This is because of its reliance on both spatial and temporal characteristics, which makes it more difficult to replicate compared to static input methods.

Voice authentication was found to be easy to use, with participants finding it convenient. This outlines the strength as an accessible authentication method, compared to more complex approaches such as gesture authentication.

Some limitations were observed during testing. The evaluation did not include attack simulations, such as replay or spoofing attacks, so conclusions regarding security are based on user perception. Participants have said that background noise can interfere when performing the voice authentication, affecting reliability in loud environments. The evaluation also focused mainly on usability rather than attack scenarios.

As a result, while voice authentication performs well in terms of user experience its security characteristics were not fully explored within this testing phase. This suggests that further evaluation is required to assess its robustness in real world conditions.

5.4 System Performance

The system showed mostly positive performance in terms of responsiveness and efficiency. Most participants agreed that the login process was reasonably quick, although some responses were rated neutral, meaning there is room for improvement.

Accuracy was also tested, with mixed feedback. While multiple participants reported that the system responded accurately to their inputs, some have said that it was not always consistent. This is particularly more targeted towards gesture and voice authentication, where feedback varies in user input and environmental factors as they can affect performance. These variations can lead to higher false rejection rates, where the actual users are denied access due to inconsistencies in input or environmental conditions. These results suggest that, while the system is functional and effective, future refinement of thresholds and detection mechanisms could improve reliability. Optimising threshold values is critical in balancing false acceptance rates (FAR) and false rejection rates (FRR), ensuring that the system remains both secure and usable.

5.5 Technical Evaluation

In addition to user based evaluation, the system was assessed in terms of technical performance and reliability during authentication.

During testing, authentication processes normally completed within a few seconds, depends on what method the user selected. Simple methods like FaceID and voice demonstrated faster completion times, while gesture authentication needed longer because of the input variability.

The system demonstrated consistent responses behaviour across multiple attempts, though some variability was observed in gesture and voice authentication because of the different inputs and the conditions of the environment. These inconsistencies may cause false rejections, in which legit users are not recognised.

In authentication systems, critical measures of performance are False Acceptance Rate (FAR) and False Rejection Rate (FRR). While these were not measured in evaluation, observations suggest that the system prioritises reducing false acceptance by requiring multiple authentication factors, which may increase false rejection in certain scenarios.

In summary, the system shows acceptable performance and reliability, with opportunities for future improvements in threshold tuning and input consistency.

5.6 Discussion

5.6.1 Overview of Findings

The results from the evaluation shows that each authentication method displays a distinct exchange between usability and security. While some methods focuses mainly on ease of use, the other methods give stronger resistance to observation attacks and attackers trying to imitate the input. This section of the document analyses these findings that relates to the project aims and existing literature.

5.6.2 Usability vs Security Trade-off

Voice authentication was deemed as the most usable method, with 70% of participants rating it as easy to use. This may be due to the natural interaction and requires users' minimal effort, making it suitable for frequent use. However, although it is more convenient it comes with the cost of reduced security, for example in scenarios where the passphrases may be overheard.

Whereas gesture based authentication demonstrates a stronger security as it relies on both spatial and temporal input characteristics. However, 50% of participants rated it as difficult, demonstrating that complex inputs impact the usability negatively. This reflects a trade-off in the design, where higher security cost the user’s convenience.

5.6.3 Evaluation of Biometrics Methods

Facial authentication in this system includes two methods, FaceID and facial motion recognition. They provide a balance of usability and perceived security. FaceID was positively rated because the user’s interaction requirement is kept to a minimum, while facial motion adds a behavioural layer, strengthening resistance to spoofing attacks.

However, these systems are dependent on conditions of the environment, like lighting and camera quality, which could impact reliability. Without advanced liveness detection mechanisms, biometric systems may be still vulnerable to spoofing attacks, outlining how important it is to combine the methods with extra authentication factors.

5.6.5 Effectiveness of Multi-modal Authentication

The findings support how effective a multi-modal authentication approach is. By combining and putting together different methods, the system does not rely on a single point of failure. For example, even if an attacker were to bypass one authentication method, other layers continue to protect.

Figure 12: Example of successful authentication showing completion of selected methods

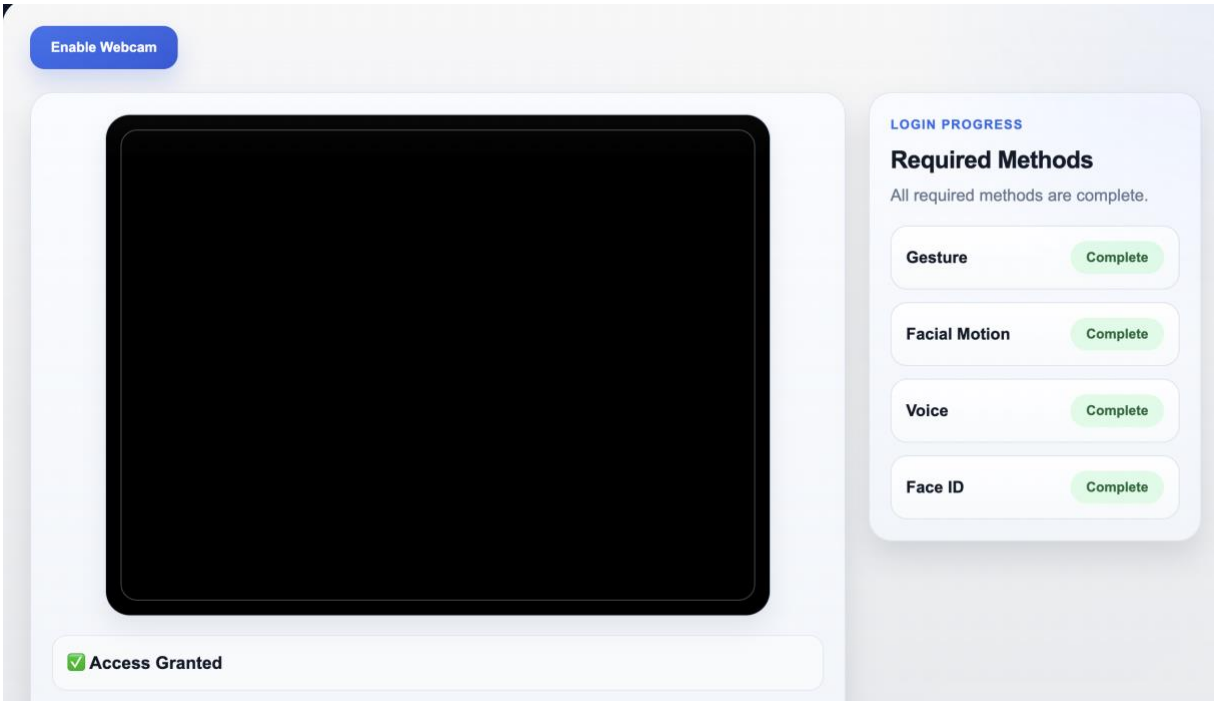


Figure 12 displays a successful authentication process in which the multiple selected methods have been completed in one session. This demonstrates how different authentication methods can operate together, giving a layered process to verify the user, enhancing security and usability.

This is effective in reducing vulnerability to shoulder surfing attacks because an attacker would need to replicate multiple types of input instead of doing one observation action.

5.6.6 Link to Literature

These findings line up with the existing research, which suggests that combining behavioural and biometric authentication methods improves the system's robustness. As noted in Jain et al. [8], multi-modal systems increase the difficulty of unauthorised access by requiring attackers to bypass multiple independent mechanisms.

5.6.7 Summary of Discussion

In summary, the discussion highlights that an authentication method by itself is not enough on its own. When combined in a multi-modal framework, the methods complement each other's strengths and weaknesses, offering a secure and adaptable system.

In conclusion, the findings demonstrate that no authentication method on its own is sufficient, reinforcing the importance of multi-modal approaches in achieving both usability and security in modern systems.

5.7 Limitations

Despite the strengths the system contains, it also has several limitations. Voice authentication is vulnerable to noisy environments and potential misuse if the passphrase is discovered. The gesture authentication was difficult for most of the users to perform consistently, mainly for complex patterns. Additionally, lighting and camera quality may affect facial performance.

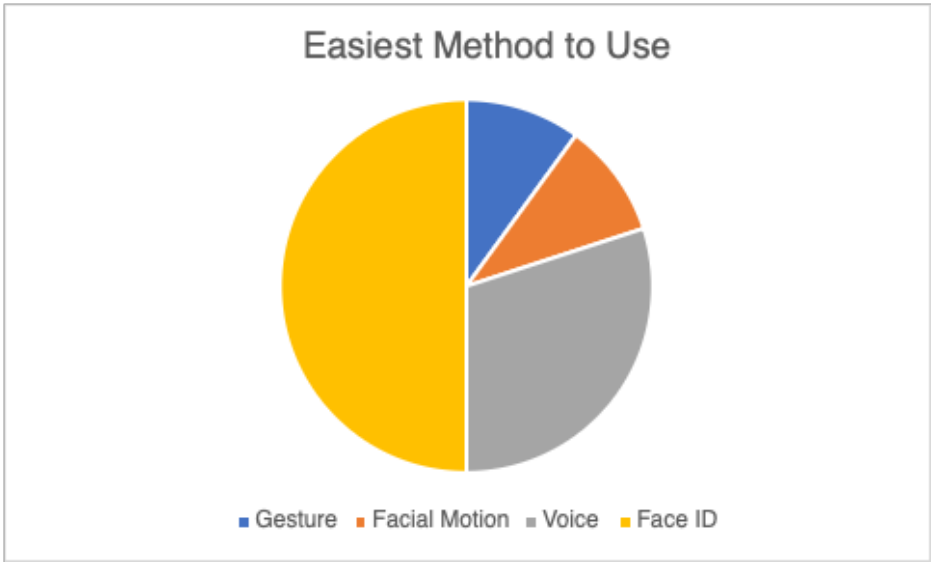
However, the evaluation sample size was small as I only got 10 participants to help with the assessment, limiting the validity of the results. A larger and more diverse group would have helped provide more reliable insights and feedback into system usability and performance.

These limitations highlight areas for improvement, particularly in robustness and adapting the system to real-world environments. Furthermore, the absence of formal attack testing means that the system's resistance to advanced threats, such as replay or spoofing attacks, cannot be fully validated within this study.

5.8 User Feedback Analysis

Participants were asked to assess which method they found easiest to use. Figure 13 illustrates the results.

Figure 13: Perception of easiest authentication method



As displayed in Figure 13, the method with the most votes is FaceID, with 50% of responses indicating it as the easiest method. This suggests biometrics methods requires less effort compared to behavioural methods. Voice authentication was second most voted as users consider it easy to use, while gesture and facial motion were the least preferred.

Another of the participants questions was which authentication method they recognise as the most secure. The results are displayed in a pie chart in Figure 14.

Figure 14: User perception on the most secure authentication method

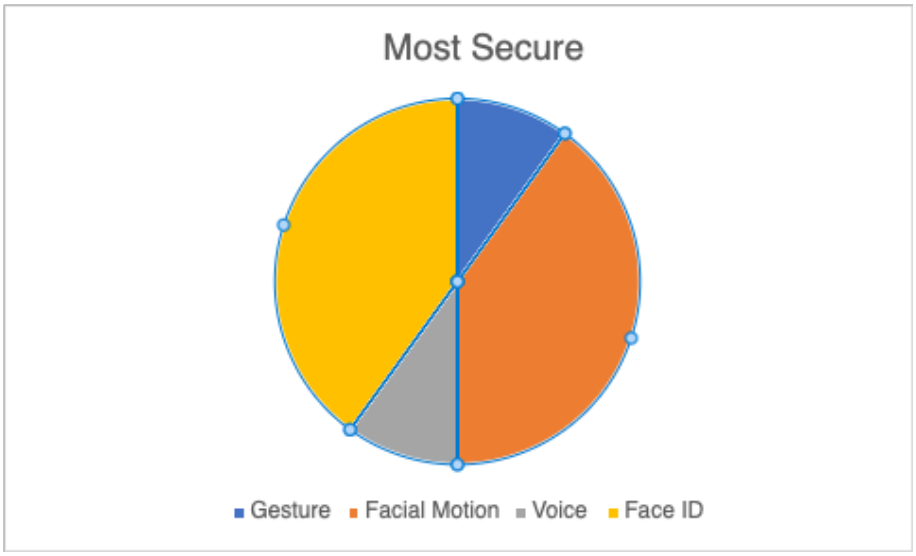
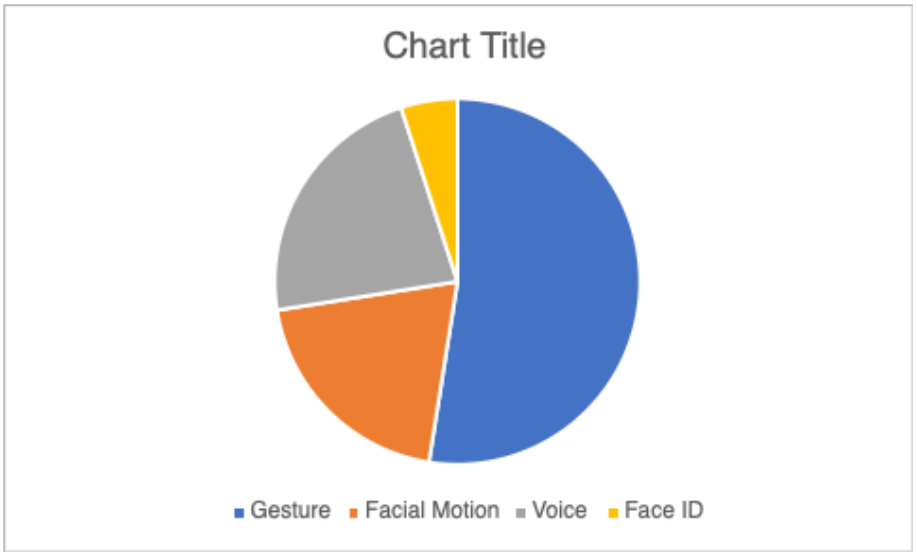


Figure 14 illustrates FaceID and facial motion authentication were seen as the most secure methods, in which they both received 40% of responses. This shows that users partner biometric and movement based authentication with stronger security. Compared to gesture and voice they were perceived as less secure.

To extend the evaluation on usability, participants were also asked about the difficulty of the authentication methods. The results are shown in Figure 15.

Figure 15: Authentication method difficulty



In figure 15, it is shown that gesture authentication had the highest rating on the difficulty rating, indicating that participants found it the most challenging to use. This aligns with feedback highlighting issues in reproducing gestures consistently. On the other hand, FaceID had the lowest rating, reinforcing its usability advantage.

5.9 Summary

Overall, the evaluation shows that the given authentication system improves both usability and perceived security compared to traditional password-based methods. While individual authentication methods have limitations, their integration in a multi-modal framework provides a more resilient and adaptable solution, demonstrating the potential of combined authentication techniques in reducing observation-based attacks such as shoulder surfing.

6. Critical Appraisal

6.1 Evaluation of Project Outcomes

The main aim of the project was to design and create a graphical authentication system that reduces the risk of shoulder surfing while maintaining usability. Overall, the system successfully met this aim by having multiple authentication methods, including gesture-based input, facial motion recognition, voice authentication, and FaceID. This shows that integrating multiple authentication methods can help reduce reliance on a single inputs, thereby improving resilience against observation attacks.

The implementation shows strong functionality, with every authentication method operating almost as intended. Gesture authentication captures dynamic input using preprocessing and sequence matching techniques, while facial motion enables behavioural verification through action based sequences. Voice authentication provides an accessible interaction method, and FaceID gives an convenient biometric solution supported by a microservice architecture.

What the evaluation results has shown was that users generally sees the system as secure and usable, specifically when compared to traditional password-based methods. However, the results also outline an exchange between security and usability across different authentication techniques. The methods that give higher security, like the gesture authentication, shows usability challenges, while more accessible methods, like the voice authentication, may give weaker protection against certain attack vectors.

6.2 Limitations and Weaknesses

Despite the strengths the system has, multiple limitations impact the overall effectiveness.

An example of a limitation is the gesture authentication usability. It provides strong behavioural security, participants sometimes reported that reproducing gestures is difficult, especially when they were complex. This highlights a key limitation of behavioural authentication systems, where increased complexity can impact

consistency and user performance in a negative manner. This indicates usability challenge that may affect adaptation in a real world scenario.

Voice authentication is easy to use but demonstrates notable security limitations. Since the system relies on a predefined passphrase, it may vulnerable if the phrase is known that can be reproduced by an attacker. Another issue is environmental factors such as background noise were observed to interfere with performance, not making it reliable in a uncontrolled settings. These limitations suggest that voice authentication is more suitable as a supplementary method rather than a main security mechanism. This strengthens the importance of multi-modal system, where weaker methods support each other by making it stronger by combining authentication factors.

FaceID authentication was reinforced with basic liveness detection by requiring the user to a blinking, improving resistance to spoofing attacks that uses static images. However, this does not fully erase the possibility of more advanced attacks, like a video of the user blinking. This demonstrates that basic liveness detection provides limited protection and may not be enough to stop against more adversarial techniques. This emphasises the need for more sophisticated liveness detection mechanisms in future implementations.

Another limitation of the evaluation is its testing scope. This resulted in conclusions regarding system security that are primarily based on perceived effectiveness rather than actual resistance to attacks, limiting the strength of security validation. The evaluation was focused mainly on usability and perceived security rather than resistance to attacks. Participants were not told to attempt shoulder surfing, meaning that the system's effectiveness against attack scenarios was not fully tested.

Furthermore, the sample size was small with only 10 participants, which may limit the general findings. While the results provide useful insights, a larger participant group would be needed for a more comprehensive evaluation. This would improve findings and provide more useful insights, especially in assessing usability across different user demographics.

6.3 Social, Ethical, and Security Considerations

The system raises important social and ethical considerations, specifically in relation to biometric data. Authentication methods such as facial recognition and voice involve the collection of sensitive personal data, which must be handled securely. Improper handling of such data could lead to privacy risks, as biometric data is sensitive and cannot be easily changed if compromised.

To address this, which is also part of the project, the system incorporates encryption for authentication templates and restricts data access through backend controls. Participants were informed about the nature of the data being collected, and their consent was given before testing, also in line with ethical research guidelines.

Multi-modal authentication systems have potential to improve security in environments where traditional passwords are vulnerable. Though, the reliance on biometric and behavioural data raises ethical considerations regarding user's consent, and data storage. They also raise concerns to privacy and accessibility. For example, some users may find some methods difficult to use due to environmental limitations.

6.4 Potential Improvements

There are improvements that could enhance the system in future work.

Gesture authentication could be refined by introducing adaptive thresholds or machine learning models to help better accuracy and reduce user effort. This could lessen false rejection rates by letting the system adapt to natural variations in user behaviour over time. Secondly, voice authentication could be strengthened by including more advanced speaker recognition techniques, instead of relying on passphrase matching. This would improve security strength by determining between users with similar voices and reducing replay attacks.

FaceID security could be improved by adding more advanced liveness detection methods, like depth-based analysis. Enhancements like that would greatly strengthen resistance to spoofing by requiring more complex user interaction. Integrating continuous authentication or behavioural monitoring could give an extra layer of security.

The evaluation process could of been widened to include attack scenarios, like a simulated shoulder surfing, to have a better assessment of the system's security performance. This would provide a more complete assessment of the system security by evaluating performance under attack conditions.

6.5 Personal Reflection

This project provided an opportunity for technical and personal development. From a technical perspective, it included working with different technologies, involving real time computer vision, backend development, and database management. The implementation of multiple authentication methods needed careful decisions to balance usability and security. This strengthened the importance of considering exchanges between system complexity, user experience, and security.

Besides technical skills, the project also helped improve problem-solving abilities, especially when addressing challenges related to accuracy, performance, and system integration. The development process pointed out the importance of iterative testing and refinement in building reliable systems. This process was essential in identifying weaknesses and improving system overall over time.

In summary, the project has given valuable experience in creating a secure authentication system and has strengthened the importance of considering both user experience with security in system design.

7. Conclusion

This project aimed to design and develop a multi-modal graphical authentication system to address the limitations of traditional passwords, especially against shoulder surfing attacks. By integrating multiple authentication methods, including gesture-based input, facial motion recognition, voice authentication, and FaceID, the system offers a more secure and flexible alternative.

The results of the evaluation demonstrates that combining authentication methods helps better usability and perceived security. Every method gives different strengths, such as gesture authentication, which displays behavioural complexity, facial motion

shows dynamic interaction, voice authentication gives accessibility, and FaceID provides convenience and efficiency. When implemented, the methods forms a layered process which helps reduce reliance one method and strengthens resistance to observation attacks.

The evaluation points out important trade-offs. Methods which provides stronger security, like gesture authentication cost their usability for the user, while the accessible methods, such as voice, give potential security limitations. Furthermore, the absence of attack simulations and small sample size, restricted to show how the system's security can be full validated.

In conclusion, the project successfully achieved its objectives and demonstrates the effectiveness of multi-modal authentication, indicating that it is practical way in improving security against shoulder surfing. The findings imply that using behavioural and biometric techniques and combining them provides a more robust and adaptable authentication solution compared to traditional methods.

For future work, could focus on improving system robustness by using advanced liveness detection, improving voice recognition, and implementing adaptive thresholding. In addition, widening the evaluation by including a larger and diverse group and simulating attack scenarios, would provide useful insight and offer a comprehensive assessment performance in real conditions.

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